

Course Introduction

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Outline

- 1 Computer Vision: What and Why
- 2 This Course: Topics, Structure, Objectives
- 3 Resources and References

What is Computer Vision?



Where is the gluestick? Find the book -
what's its full title?

Credit: Bharath Kishore, Flickr CC License



What is wrong with this image?

Credit: Erik Johansson

What is Computer Vision?



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Can a machine answer the above questions?

What is Computer Vision?

Computer Vision

- A field that seeks to automate and endow a computing framework with the ability to interpret images the way humans do.
- A sub-topic of Artificial Intelligence.

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Other Definitions

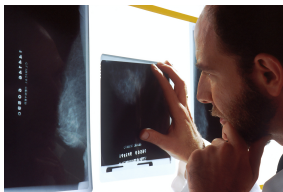
- “the construction of explicit, meaningful descriptions of physical objects from images” (Ballard & Brown, 1982)
- “computing properties of the 3D world from one or more digital images” (Trucco & Verri, 1998)
- “to make useful decisions about real physical objects and scenes based on sensed images” (Sockman & Shapiro, 2001)

Why? Applications of Computer Vision



Autonomous Vehicles

Credit: smoothgrover22, Flickr CC License



Medical Imaging

Credit: National Cancer Institute



Surveillance

Credit: Yeong Nam, Flickr CC License



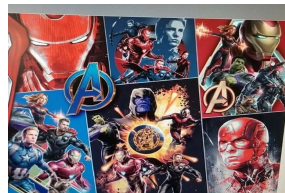
Human-Computer Interaction

Credit: Vancouver Film School



Factory Automation

Credit: KUKA Roboter GmbH, Bachmann



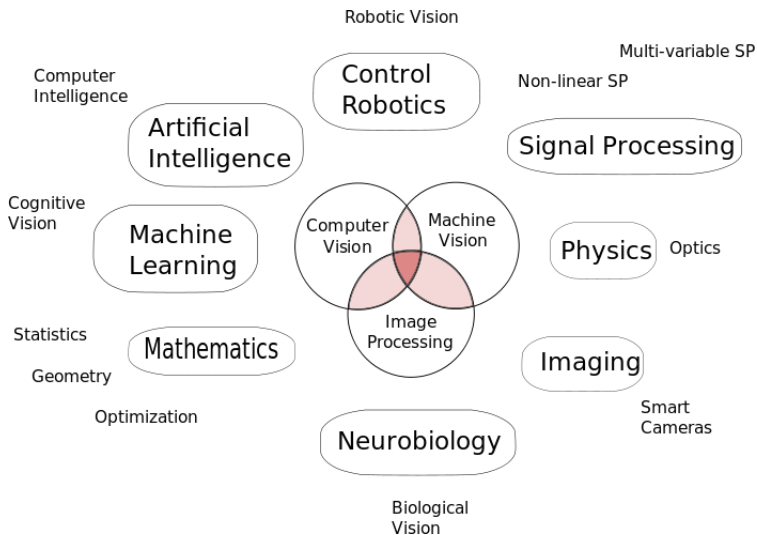
Visual Effects

Credit: AntMan3001, Flickr CC License

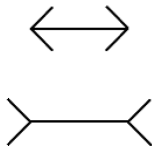
Applications of Computer Vision: More...

- **Retail and Retail Security** ([Amazon Go](#), [Virtual Try-on](#), [StopLift](#))
- **Healthcare** ([Blood Loss Detector](#), [DermLens](#))
- **Agriculture** ([SlantRange](#), [Cainthus](#) - [Livestock facial recognition](#))
- **Banking and Finance** ([Mobile Deposit](#), [Insurance Risk Profiling](#))
- **Remote Sensing** ([Land Use Understanding](#), [Forestry Modeling](#))
- **Structural Health Monitoring** ([Oilwell Inspection](#), [Drone-based Bridge Inspection](#) and [3D Reconstruction](#))
- **Document Understanding** ([Optical Character Recognition](#), [Robotic Process Automation](#))
- **Tele- and Social Media** ([Image Understanding](#), [Brand Exposure Analytics](#))
- **Augmented Reality** ([TechSee Visual Support](#), [Warehouse and Enterprise Management](#))

Perspectives of Study



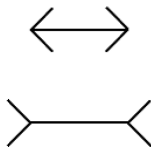
Why is it hard?¹



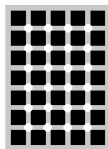
Müller-Lyer illusion: Which line is longer?

¹Credit: Szeliski, Computer Vision: Algorithms and Applications, 2010

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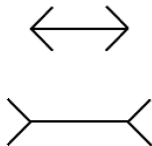
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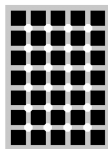
Variation of Hermann grid illusion: What do you see at the intersections?

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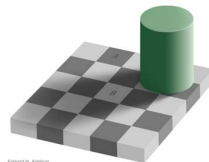
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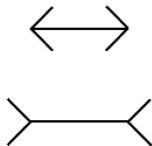
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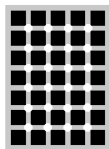
Adelson's brightness constancy illusion:
Which is brighter, A or B?

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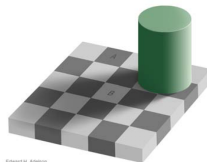
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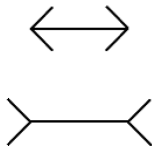
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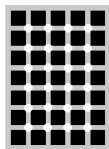
Count the red Xs in both figures, which is harder?

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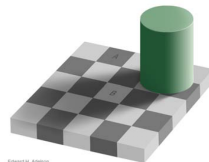
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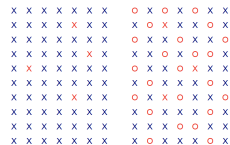


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Why is it hard?

- Many practical use cases are **inverse model** applications
 - No knowledge of how an image was taken or camera parameters - but need to model the real world in which picture/video was taken (shape, lighting, color, objects, interactions). $\implies$ Need to almost always model from incomplete/partial noisy information
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 - Forward models* are used in physics (radiometry, optics, and sensor design) and in computer graphics
- High-dimensional data $\Rightarrow$ heavy computational requirements
- Computer vision is **AI-complete**



IN CS, IT CAN BE HARD TO EXPLAIN THE DIFFERENCE BETWEEN THE EASY AND THE VIRTUALLY IMPOSSIBLE.

Credit: [Anish Chopra, Medium.com](#)

Why is it hard?

- No complete models of the human visual system exist
 - Existing models largely related to subsystems, not holistic
 - What is perceived, and what is cognized? When is an object important for a task, and when is the context important?

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- No complete models of the human visual system exist
 - Existing models largely related to subsystems, not holistic
 - What is perceived, and what is cognized? When is an object important for a task, and when is the context important?
- Verifiability of mathematical/physical models non-trivial
 - How should similarity/dissimilarity between representations be defined? Is this a distance metric? Do all images follow such a distance metric?
 - How would a manipulation (counterfactual) in a given (potentially noisy) environment behave, w.r.t. the captured image/video? Can a physical model capture this?



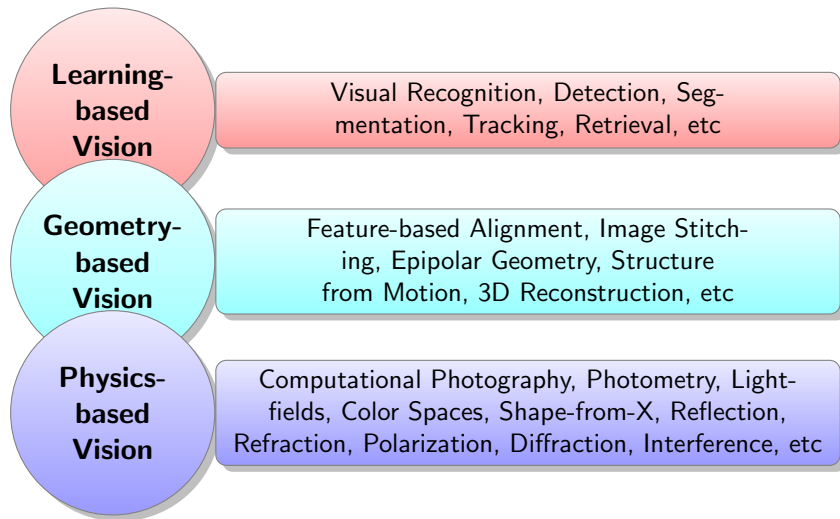
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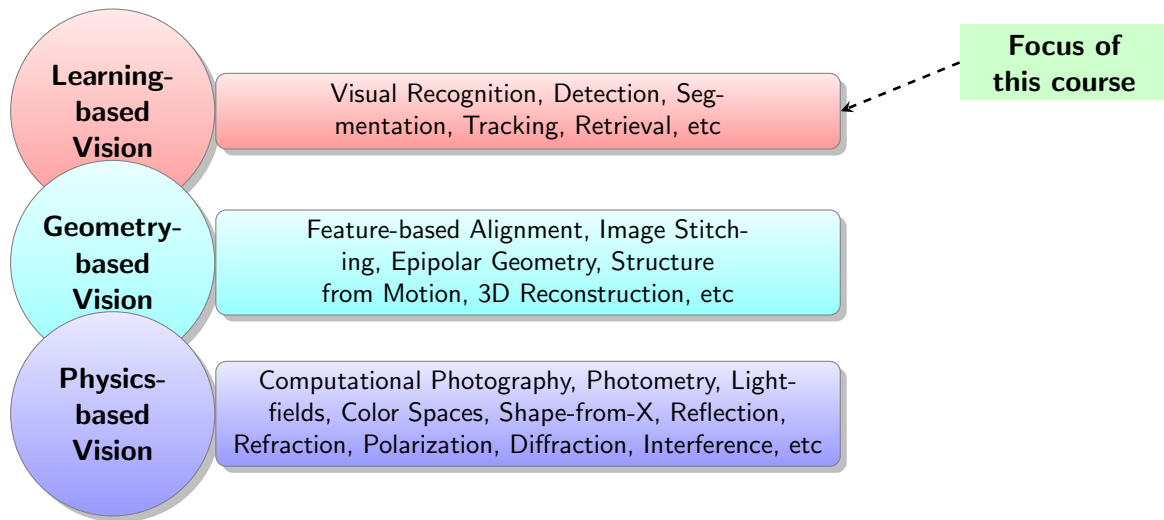
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Computer Vision: Topics



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Course Topics

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- Image Formation, Linear Filtering
- Edges, Blobs, Features
- Visual Descriptors

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- CNN Architectures, Variants and Visualizing CNNs
- CNNs for Detection and Segmentation

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- Video Understanding using CNNs and RNNs
- Attention in Vision Models
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- Generative Adversarial Networks
- Variational Autoencoders
- Diffusion Models

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Segment 5: Recent and Advanced Topics

- Self-supervised Learning, Contrastive Learning
- Vision-Language Models
- Foundation Models for Vision
- Few-shot, Zero-shot Learning
- Robustness, Model Compression, Neural Architecture Search
- Recent Vision-Language Models

Course Eligibility

Theory

- Completion of a basic course in Machine Learning
- (Strongly recommended, not mandatory) Completion of a course in Deep Learning, or exposure to topics in neural networks
- Knowledge of basics in probability, linear algebra, and calculus

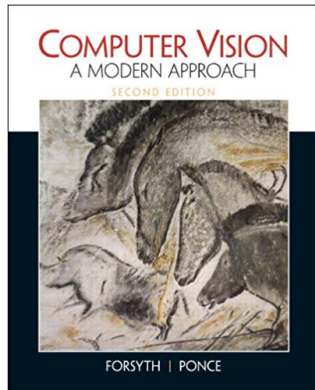
Programming

- Experience of programming in Python
- Knowledge of a deep learning framework (PyTorch or TensorFlow) highly recommended

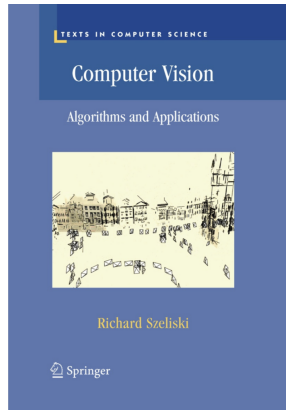
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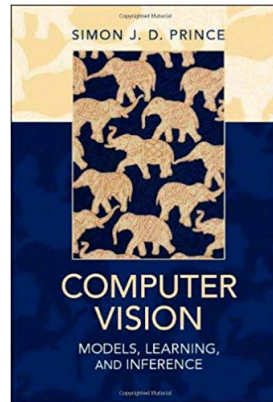
Traditional Computer Vision: References



[Book website](#)

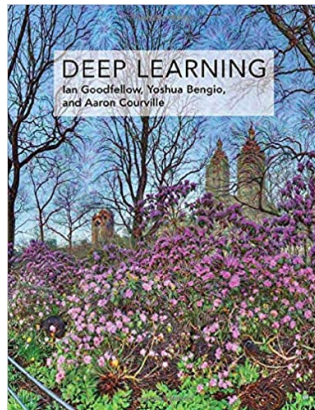


[Book website](#)



[Book website](#)

Deep Learning: References



Book website

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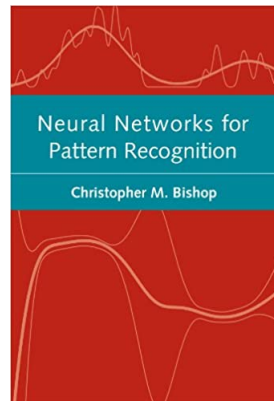
Neural Networks and Deep Learning

Neural Networks and Deep Learning is a free online book. The book will teach you about:

- Neural networks, a beautiful biologically-inspired programming paradigm which enables a computer to learn from observational data
- Deep learning, a powerful set of techniques for learning in neural networks

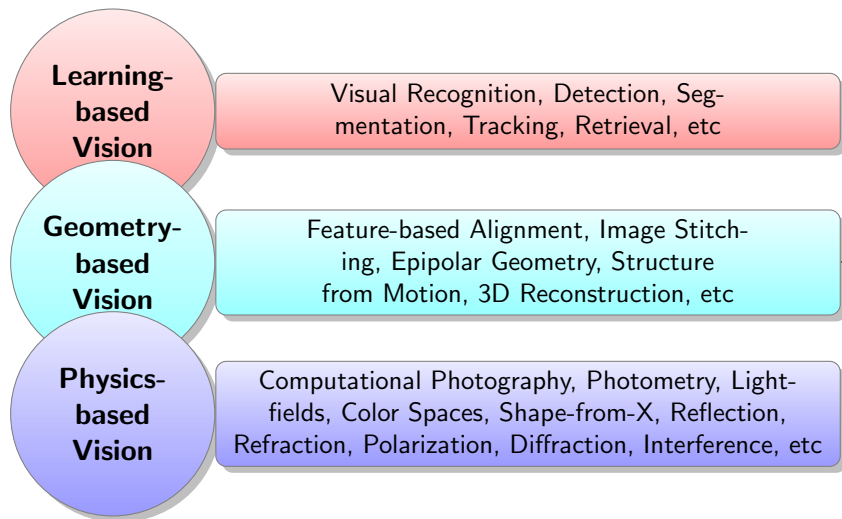
Neural networks and deep learning currently provide the best solutions to many problems in image recognition, speech recognition, and natural language processing. This book will teach you many of the core concepts behind neural networks and deep learning.

A nice, short online book
by Michael Nielsen



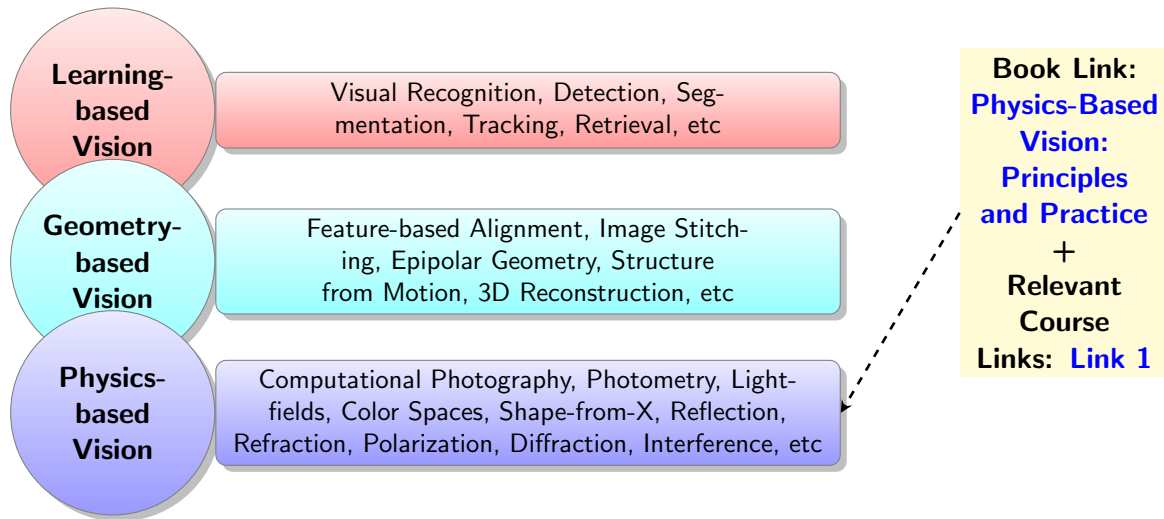
Book website

Want to Learn Other Topics?



Book Link:
**Multiple View
Geometry in
Computer
Vision**
+
**Relevant
Course
Links: [Link
1](#), [NPTEL
Course](#), [See
Weeks 2-5](#)**

Want to Learn Other Topics?



Homework!

- Go through all links on the Applications of Computer Vision slide (Slide 6) - they are interesting views/reads!

Acknowledgements and Disclaimers

- We are grateful to the deep learning/computer vision courses and their contents that are publicly available online. Wherever possible and relevant, these sources have been cited. If you notice an oversight, please let us know, and we will be glad to acknowledge.
- Any errors in the material are our own. Please point out such issues, and we will be glad to rectify.
- To the extent possible, all images used in these materials have been chosen from free stock photos to avoid any copyright violations. If you notice an oversight, please let us know.